

Morphological Processes in Nupe Numeral System

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Abstract

Numerals are signs, marks, symbols or words used to represent a number. This study examines the morphological derivation of Nupe numerals. The study adopts a descriptive and analytical research design involving the description and analysis of qualitative data gathered through both primary and secondary sources, with oral interview as the main source of data collection from native speakers of Nupe language from Nupe speaking areas of Niger State, Nigeria. The lexical phonology theory was the theoretical framework adopted for the study. The data collected were analyzed descriptively. The findings from the study reveals that Nupe numerals are derived through the use of additive, subtractive and multiplication strategies manifested through morphological process such as affixation, compounding and reduplication. The study concludes that adequate knowledge of these morphological processes is imperative for competence and proficiency in the Nupe counting system. The study recommends further research on Nupe numeral system and language; while efforts to record and document traditional counting systems of minority languages are also suggested.

Keywords: Morphology, Affixation, Compounding, Reduplication, Agglutination, Numerals, and Nupe.

Introduction

Numerals are aspects of language studies which researchers, scholars and linguists pay less attention to. This neglect could be connected to the impression that there is little or nothing else to say about it. Little do we know that there is much to say about this 'little'?. It is argued that, part of our linguistic knowledge is contained in the ability to express our ideas in our mother tongues. Thus, our linguistic knowledge remains insufficient if we are unable to count fluently, analyze how the counting is done, and understand manipulations therein. However, the traditional numeral/counting system, in any given speech community, constitutes one of the sociolinguistic factors that make up the distinctiveness and individuality of that speech community as against other speech communities (Babarinde, 2013). It is argued in this study that the numerals or counting system is a product of human intelligence and limitless of his capacity for in-depth thoughtfulness. According to Isyaku (2012), Nupe numeral system originated from counting of toes as well as fingers in reckoning numbers and it is not uncommon in African languages such as Yoruba, Igbo, Hausa and a host of others. Thus, the fact that every speech community has its own number words used for counting attests the claim that language permeates every aspect of human activities.

The Nupe language is one of the minority languages in Nigeria, and as it is the case with most minority languages in the country, Nupe is insufficiently studied. It cannot be compared with major languages such as Igbo, Hausa and Yoruba in terms of literature and research. There is paucity of linguistic studies on the numerical system of the language for both academic and pedagogic purposes. Although there are some studies on Nupe, they were mostly socio-linguistics inclined. Previous works on the numeral system of Nupe such as Alkali (2010), Isyaku (2012) and Isyaku (2016) only presented the numbers in their order of occurrence and the derivation process

without adequate information, identification, analysis and description of the structure of the morphemes and other linguistic units such as root words, affixes and word classes that characterize the Nupe numerals. The linguistic components of the numerical system of language, especially the morphological structures have not received proper and considerable attention from linguists and researchers over the years.

Morphology refers to “the study and analysis of the structure, forms and classes of words” (Hartmann, and Stork, 1972: 146). It is the scientific study of forms and structure of words in a language (Sunubi, 2016). Morphology is the study of the structure of words. It is that branch of linguistics that studies the internal structure of words and how words are formed in a language. Thus, morphology accounts for word formation in languages. As a branch of linguistics, morphology deals with the structure and formation of words in a language. It focuses on the rules and patterns that govern how words are created, including the study of prefixes, suffixes, root words, and how they combine to form words. Morphology also explores the different forms that words can take, such as tense, number, and gender in many languages. It plays a crucial role in understanding the building blocks of a language’s vocabulary and how words are related to one another. Thus, the study of morphology is important to the general study of language because “of all the distinct aspects of language, morphology is the most deeply intertwined with the others” (Aronoff & Fundeman, 2005:22).

This study therefore examines the morphological derivation of numerals in Nupe. The primary focus of this study is to contribute to the documentation and description of this most vulnerable aspect of Nupe grammar by providing a detailed description of the morphology of Nupe numerals. The study examines the morphological structures that characterize the derivation of numerals (cardinal and ordinal) in the language.

The name Nupe refers to the land, people and language. The people are called ‘Nufawa’ in Hausa, ‘Takpa’ in Yoruba and recorded as ‘Tappa’ in Migeod (1913:340). According to Ismaila (2002), Nupe is a word used in describing people living in the confluence of Niger and Kaduna rivers. Thus, Nupe is the language of the people that live in the area between the Niger and Kaduna rivers.

According to Greenberg (1963), Nupe belongs to the Kwa-languages. He classifies Nupe under kwa sub-group ‘D’ linguistic phylum comprising Gwari/Gbari (Gbagyi), Egbira and Gade. Hoffmann (1976) refers to Nupe, Gbagyi and Gade as the ‘Niger-Kaduna group’. Bennett and Sterk (1977) label the group ‘central - Niger languages’. Williamson (1987) calls the group ‘Nupoid’. The name Nupoid was first introduced by Hyman (1972:175) and commonly used in the literature to refer to the group of languages whose principal representatives are Nupe, Gbari, Gade and Ebira (formerly Niger-Kaduna). Thus, the Nupe language belongs to the Nupoid branch of the Benue-Congo group of languages. Blench (1989) and Blench (2013) classify Nupe as a Benue-Congo language of the Nupoid phylum.

The population of Nupe as at 1963 was 650,000 and by 1986 the number grew to 1,314,000 (Jibril, 1990: 111) and has continued to increase. With an estimated 800,000 L1 speakers in 1990 and about 200,000 L2 speakers by 1999; Nupe language is spoken as a mother tongue by at least about two million people (Hair, 1969:70). Nupe is considered to be one of the minor languages in Nigeria (next in rank to the three major languages of Hausa, Yoruba and Igbo). Nupe is spoken in Niger State, Kwara (North), Kogi State and the Federal Capital Territory (FCT) (Musa & Oyebade, 2018).

Literature Review

Numerals

Generally, numerals can be described as a sign, mark or symbol used to represent a number; and even though, numeral system is a written system for expressing numbers, in linguistics, numerals are specific words in a natural language that represent numbers (Obikudo, 2013:27). Numerals are among one of the first things we learn when learning a new language. The acquisition of numeracy starts from the counting base of the language that depends on the structure of a language (Mbah & Uzoigwe, 2013:73). Numerals are uniform because in many natural languages they comprise simple and complex expressions. Simple numerals are the easiest conceivable set of numerical expression in a language (e.g. one, two, three, four and so on.). However, the simple numbers are preferred for having a 'cyclic' pattern of atoms that cannot be separated and also morphologically derived to form the complex elements of numerals that have the highest potential to make a continuous counting in each language. These can be in the form of addition, subtraction and multiplication. Numerical system is therefore, the pattern of naming or presentation of symbols, or signs to represent numbers corresponding to numerical quantities, values, or measurements. Numerals often come together with a set of rules for representing larger numbers. They provide a concise and standardized way to express numerical quantities effectively. However, there are three types of numerals: Ordinals, Cardinals and Distributive numerals used in the numerical system of languages; and the systems depend on some rules on how to count things by the native speakers of a language (Sanusi & Yusuf, 2018).

1. **Ordinal Numerals:** These are natural numbers that are used to measure the "length" or "shape" of a set of construction. In linguistics, ordinal numbers are used to represent the *position* or *rank* in a sequential order. The order may be of size, importance or chronology. Example of ordinals series in English are also: first, second, third, fourth, fifth, sixth, seventh, eighth, ninth, tenth and so on.; and may be written in English with numerals and letters of suffixes (that is, 1st, 2nd, 3rd, 4th, and so on). These are used in a variety of rankings, including *time* (Example the first hour, the second week, the third month of a year, second round, first quarter, second semester and so on.), *space* (Example the first right, the second floor and so on.), *position* (Example. first winner, second person, third person and so on.) and *quality* (first class ticket, first choice, and so on.). In Latin, the ordinal series have been specified as: primary, secondary, tertiary, quaternary and so on (Sanusi & Yusuf, 2018). However, ordinal numerals describe the position of objects in a sequential order apart from the first and last ordinal numerals (Obikudo, 2013:34).
2. **Cardinal Numerals:** Cardinal numbers are a generalization of the natural numbers used to measure the cardinality (size) of a set. The cardinality of a finite set is a natural number of an element to a set that transfinite the cardinal numbers to describe the sizes of infinite sets. Thus, cardinals are considered basic in form used for counting and in describing quantity (Sanusi & Yusuf, 2018). They are used to count, as follows: one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen etc. However, cardinal numerals can be divided into *simple* and *complex* numerals by considering the short and long words that represent the numbers in languages when the speakers of a given language come across to a problem related to their numerical expressions.
 - i. **Simple Cardinals:** The simple cardinals are the easiest conceivable set of numerical expressions in a language, which are in morphemic structure and arbitrary found with

phonological shape. They are numerals that are not derived; that is, numerals whose forms cannot be broken down into indefinable meaningful morphemes where the base numbers of each language are simply representing the basic cardinals (Mbah & Uzoigwe, 2013:73).

- ii. **Complex Cardinals:** The complex cardinals are morphologically derived from the simple elements of the numeral systems that used to form the other numerals for counting in each language after its ten-base and the arithmetic operations of addition, subtraction and multiplication are also considered to be the processes of having the complex numerical expressions. Thus, the complex cardinals are also morphological account of numerals which are considered more complex structures.
3. **Distributive Numerals:** Distributive numerals are basically set of numerals that are derived from cardinal numerals via either partial or complete reduplication. According to Ajiboye (2013:2) distributive of numerals occur in two forms; the base form and the multiple forms which undergo reduplication to form expressions of universal quantification, and it may be full or partial reduplication. However, distributive numerals also express a group of the number specified and describe an alternating order either by partial or total reduplication of cardinal numeral.

Morphology

According to Crystal (1980:232-233), morphology is a branch of grammar that studies the structure or the form of words, particularly through the use of morpheme. Katamba (1989) and Booij (2007) posit that, morphology suggests an aspect of linguistic study concerned with the study of word structure and the relationship between forms and meaning. It is primarily concerned with how words are formed. Put differently, morphology is the identification, analysis and description of the structure of a given language's morphemes and other linguistic units like root words, affixes, intonation, stress or implied context (Aronoff & Fudeman, 2011).

Generally, morphology is divided into two fields: The study of inflectional (inflectional morphology), and the study of word formation (lexical or derivational morphology) where the emphasis is on technique of analyzing words into morpheme. According to Bauer (1983:33), morphology studies the internal structure of word form. In morphology, the analysis divides word form into constituent formatives (most of which are morphs in the form of roots or affixes), and attempts to explain the occurrence of each formative. Hence, morphology can be divided into two main branches, namely inflectional morphology and word formation (also called lexical morphology). Inflectional morphology studies various forms of lexemes, while word formation studies new lexemes from certain bases. Word formation is how new words are created. There are different ways words can be formed, called morphological processes. Derivation which is concerned with the formation of new lexemes through affixation; is one process where a new word is formed by adding a prefix or suffix to an existing word. On the other hand, compounding is concerned with the formation of new lexemes from two or more potential stems. Derivation can also sometimes to be divided into class maintaining derivation and class changing derivation. Class maintaining derivation is the derivation of new lexeme whose class is the same as base from which the lexeme is formed, while class changing derivation produces lexemes whose class is different from its base. Compounding is usually divided according to the class or category of words of the compound words that are produce into compound noun, compound verb, compound adjective (Sunubi, 2016). Other morphological processes to word formation include:

- i. **Reduplication:** This when part or all of a word is repeated to form a new word;

- ii. **Blending:** This is when you take parts of two or more words and combine them into a new single word;
- iii. **Clipping:** This is a process where parts of words are cut off or deleted to make shortened forms of the original word;
- iv. **Back-formation:** This is when a new word is created by seemingly removing a suffix from an existing word;
- v. **Functional Shift (conversion or category change):** This is a change in the part of speech or grammatical category;
- vi. **Extension of Word Formation Rules:** Word formation rules can sometimes be extended or stretched to treat parts of words as if they are meaningful morphemes (units of meaning), even when they are not separate morphemes;
- vii. **Onomatopoeia:** This denotes the creation of lexical units by the imitation of sounds in nature (or recently in technology);
- viii. **Coining (Coinage):** This is when a completely new word or morpheme is created that is unrelated to any existing words;
- ix. **Borrowing:** All languages need to continually expand their vocabulary over time. One way is through lexical borrowing; and so on.

However, it is argued that the morphology of numerals is divided into two: basic and derived. The basic numerals refer to the traditional basic sets of numerals; while the derived numerals are the set of numerals derived through morphological processes such as derivation, compounding, reduplication, borrowing and so on. This study is to establish the underlying morphological processes that are inherent in the numerals of Nupe language.

Empirical Review

There are quite a number of empirical studies on numerals and numeral systems of African and European languages. Some of the most recent studies are reviewed as they relate to the morphological derivation of numerals of some Nigerian languages.

Abdullahi (2017) examined the Hausa and Yorùbá numerical systems from a morphological productivity approach. The study discussed the morphological productivity of Hausa and Yorùbá numerals based on word-based morphology theory where some morphological processes such as clipping, blending, compounding and reduplication were analyzed. Observation technique was used in the process of data collection. The findings indicate that the numerical system of Hausa and Yoruba are a product of interplay of morphological and phonological processes that are employed to derive various numerical entities of different quantities and magnitudes. Lexical items which represent the base numbers are characterized by various possible combinations to the cardinal numerical resources in order to build up an increasing system of counting by means of productive mechanism. The Word-Formation Rules perform a unique operation on the word base to produce a new word where both the new word and the base share a morphological and semantic relationship. The study concluded that both Hausa and Yorùbá share some similarities and dissimilarities in numerical systems. Thus, the relationship identified in numeracy may not be all as there are possible correlations in other bases or contexts such as syntactic, semantic, morphophonological, morphosyntactic and morphosemantic analysis. Sanusi and Yusuf (2018) examined the morphological properties of Bātōnū numerals. The study had the objectives to show a descriptive analysis of the derivational processes of the numeral system, to show the relevant

constraints or imperatives for well-formedness (addition, multiplication, subtraction and conjunction), to examine the representation of number in quality and amount, in line with the system of forming complex numbers and counting operations, finally, to determine the patterns and also the agreed method of counting in Bātōnū. The study adopted Mohanan's (1982) and Kiparsky's (1982a) lexical phonological and morphological theories based on their problem-solving effectiveness and their reliability in handling African language data. Data for the study were generated from both primary and secondary sources. The informant method was adopted in collecting data, through the use of frame technique. The study found that Bātōnū employed subtractive mechanism in the formation of numeral system. The language uses productively the additive element *kà* meaning 'and' in adding other base numerals. The study also found that money counting system in Bātōnū employs the mathematical operations of addition, subtraction and multiplication in order to achieve simplicity in the counting system. Jimeyough and Ishima (2023) explored the derivational processes in the Tiv numeral system. The investigation focused on how the phonemes and morphemes play significant role through variations in the word building processes of the numeral system.

The study relies on three major sources in gathering the data for the descriptive analysis namely: compilation of a list of Tiv numbers, the researcher's intuitive knowledge of the language in addition to his training as a linguist, and scholarly works on Tiv numerals in textbooks and journals available. The study employs the lexical phonology theory to back up its claims on derivational processes of Tiv numeral system. From the analysis of data, the study found that, Tiv language has complexity of deriving, especially non-basic numeral involves addition and multiplication in addition to the grammatical processes; drawing from phonological and morphological processes such as compounding, affixation and reduplication which are compiled to yield several numeral forms in the language. Hence, the study concluded that, Tiv language could be said to belong to the group of languages which have a vigesimal numeral system. This is because the language employs a numeral structure where counting is done majorly in multiples after the basic numeral of one to ten. However, the study acknowledges that it is not an exhaustive account of the numeral system in Tiv as it is limited to derivational processes. However, the literature gap exposed from the review of previous studies were not holistic, particularly in terms of exploring the morphological derivation of numerals in Nupe language. Hence, the present study fills the gap in literature as it examines the morphological features that characterize the Nupe numerals.

Theoretical Framework

This study is anchored on the lexical phonology theory developed by Mohanan (1982/1987) and Kiparsky (1982a). The theory accounts for the interactions of morphology and phonology in a language, including the numerical system of the language. With the emphasis on morpho phonology, it is a theory in which phonological and morphological rules are brought together within a single framework. It is an approach to phonology that accounts for the interactions of morphology and phonology in the word building process and the approach is based on the insight that much of phonology operates in language. Word formation rules guide the formation of words. These rules combine morphemes to form new words and are also called morphophonological rules. Goldsmith (1990:20) states that, "lexical phonology is arrived at by the concatenation of the morphophonemic rules and the purely phonological rules of the pre-generative years". Hence, the lexical phonology theory is proposed to account for the interrelationship between the two levels of

linguistics, phonology and morphology. The theory of lexical phonology is preferred over several morphological theories such as word formation theory, generative phonology theory, and construction morphology theory. This is due to the suitability of the theory to the subject matter of investigation in this study. Lexical phonology theory, according to Kenstowicz (1994: 20), receives generative interpretation of the relations between phonology and morphology. Hence, the choice of the theory is informed by its problem-solving efficiency. The lexical phonology theory, which exploits the interrelationship between morphology and phonology is capable of explaining phenomena in African languages (and indeed, numeral system in Nupe), which were hitherto regarded as 'complex' or intractable.

Methodology

This study is a descriptive study and the qualitative research design was adopted for the study. Data for this work were collected using primary and secondary sources. The primary sources included the native speakers of Nupe language from Nupe speaking areas of Niger State, Nigeria. Oral interviews were conducted with thirty (30) elderly members from the six (6) local government areas: Agaie, Bida, Edati, Gbako, Katcha and Lavun; who were randomly selected as respondents based on their deep experience and knowledge of Nupe sociolinguistics. The observations and intuition of the researcher as a native speaker were also very relevant for the work. The secondary sources on the other hand, included: scholarly works on numerals and numerical systems of African and Nigerian languages in textbooks, journals and the internet, as well as some published and unpublished materials on the numerical system of the Nupe language. However, the data collected were analyzed descriptively, based on the numerals identified by native speakers of the language and those collected from secondary sources coupled with the researcher's intuitive knowledge as a native speaker of the language. The lexical phonology theory was the tool for analysis in this study. The theory was used to exploit the interrelationship between morphology and phonology in explaining the morphological derivation of numerals in Nupe. Hence, the analysis focused on the morphological derivation and features that underlie the Nupe numerals.

Data Presentation and Analysis

Data for the study is presented and analysed based on the objectives of the study which is to analyze the morphological derivation and features of the numerals (cardinal and ordinal) in Nupe. This has to do with the identification, analysis, and description of the structure of the morphemes and other linguistic units such as root words, affixes and word classes that characterize the Nupe numerals. Thus, Nupe numerals exhibit the following morphological derivations:

Affixation

Affixation is known as *awuba* in Nupe. *Awuba* (Affixation) is extensively used in Nupe numeric tradition. Its root word *wuba* (fix/affix), is a verb which simply means to attach, fuse or stick. It allows adding of letters before and after a root or base word to give another meaning or grammatical stance to the new word. In Nupe numeration, morphological affixes have helped to produce countless higher numbers. It organizes the standard root or base numbers in the formation of numbers in tens, hundreds, thousands, millions and billions.

In Nupe numeration, there are obvious relationship between bases and their multipliers. The bases are the root words or free morphemes, while the times or multipliers are standard affixes used in the formation of numbers progressively. One interesting fact is that as the numbers grow

bigger and bigger, there is a level of flexibility as to the affixing of bases and times which can take the form of prefixes or suffixes progressively. Hence, there are four (4) types of affixes in Nupe generally: *wubaca* (prefix), *wubati* (suffix), *wubayezuma* (circumfix) and *wubatacin* (infix).

Affixation in Nupe Cardinal Numerals

Wubaca (Prefix)

This is the numeric affix added before a root word to form higher numbers. Examples of numeric prefixes in the Nupe cardinal numerals are: *gú* and *gba*.

The prefix *gú* affixed with the numeric multiples are seen in the following examples:

1.

<i>gú</i>	<i>bà</i>	=	<i>gúbà</i>	(two)
<i>gú</i>	<i>tá</i>	=	<i>gútá</i>	(three)
<i>gú</i>	<i>ni</i>	=	<i>gúni</i>	(four)
<i>gú</i>	<i>tsùn</i>	=	<i>gútsùn</i>	(five)
<i>gú</i>	<i>tswanyi</i>	=	<i>gútswanyi</i>	(six)
<i>gú</i>	<i>twabà</i>	=	<i>gútwabà</i>	(seven)
<i>gú</i>	<i>tòtá</i>	=	<i>gútòtá</i>	(eight)
<i>gú</i>	<i>twani</i>	=	<i>gútwani</i>	(nine)
<i>gu</i>	<i>wò</i>	=	<i>guwò</i>	(ten)

Also, there are instances where *gba* ‘two thousand’ is used as prefix to numeric root words to generate higher numbers as seen in the following example:

2.

<i>gba</i>	<i>shi</i>	=	<i>gbashi</i>	(forty thousand)
<i>gba</i>	<i>kpákó</i>	=	<i>gbakpákó</i>	(four hundred thousand)

Wubati (Suffix)

These are the affixing of the numeric numbers to form any number in Nupe numeration. Hence, the numeric suffixes in the Nupe numeric system are the following multiples: *bà*, *tá*, *ni*, *tsun*, *tswanyi*, *twabà*, and *tòtá* and the three numeric root words are *eshi* (*shi*) ‘twenty’, *kpákó* (*kpa* or *kpau*) ‘two hundred’ and *egba* (*gba* or *gbau*) ‘two thousand’. However, when the numeric suffixes are fixed to the numeric root words, they produce new higher numbers.

As shown in the example below, when the root word *eshi* (*shi*) ‘twenty’ is used with the multiple suffixes *bà*, *tá*, *ni*, *tsun*, *tswanyi*, *twabà*, and *tòtá*, it produces the following:

3.

<i>shi</i>	<i>bà</i>	=	<i>shibà</i>	(forty)
<i>shi</i>	<i>tá</i>	=	<i>shitá</i>	(sixty)
<i>shi</i>	<i>ni</i>	=	<i>shini</i>	(eighty)
<i>shi</i>	<i>tsun</i>	=	<i>shitsun</i>	(one hundred)
<i>shi</i>	<i>tswanyi</i>	=	<i>shitswanyi</i>	(one hundred & twenty)
<i>shi</i>	<i>twabà</i>	=	<i>shitwabà</i>	(one hundred & forty)
<i>shi</i>	<i>tòtá</i>	=	<i>shitòtá</i>	(one hundred & sixty)

The use of the numeric root word *kpákó* (*kpa* or *kpau*) which has ‘two hundred’ value in Nupe numeration generates new higher numbers with the affixing of the multiple suffixes as seen in the examples below:

4.

kpau bà = *kpaubà* (four hundred)
kpau ta = *kpauta* (six hundred)
kpau ni = *kpauni* (eight hundred)
kpau tsun = *kpautsùn* (one thousand)
kpau tswanyi = *kpautswanyi* (one thousand two hundred)
kpau twabà = *kpautwabà* (one thousand four hundred)
kpau tòtá = *kpautòtá* (one thousand six hundred)
kpau twani = *kpautwani* (one thousand eight hundred)

The numeric base or root word *egba* (*gba* or *gbau*) has ‘two thousand’ value. Therefore, when any of the numeric multiple suffixes is affixed to it, it will produce a higher number as in the following:

5.

gbau bà = *gbaubà* (four thousand)
gbau tá = *gbautá* (six thousand)
gbau ni = *gbauni* (eight thousand)
gbau tsun = *gbautsun* (ten thousand)
gbau tswanyi = *gbautswanyi* (twelve thousand)
gbau twabà = *gbautwabà* (fourteen thousand)
gbau tòtá = *gbautòtá* (sixteen thousand)
gbau twani = *gbautwani* (eighteen thousand)
gbau wò = *gbauwò* (twenty thousand)

Wubayezuma (Circumfix)

This is another interesting circumstance where prefixes and suffixes are added to a root numeric word simultaneously to give new compound numbers.

When *eshi* (*shi*) ‘twenty’ is considered as the root word, and *egba* (*gba* or *gbau*) ‘two thousand’ prefixed to it, while the multipliers as the suffixes to the same word, they will produce the following compound numbers:

6.

<i>gba</i>	<i>shi</i>	<i>bà</i>	= <i>gbashibà</i> (eighty thousand)
<i>gba</i>	<i>shi</i>	<i>tá</i>	= <i>gbashitá</i> (one hundred and twenty thousand)
<i>gba</i>	<i>shi</i>	<i>ni</i>	= <i>gbashini</i> (one hundred and sixty thousand)
<i>gba</i>	<i>shi</i>	<i>tsun</i>	= <i>gbashitsun</i> (two hundred thousand)
<i>gba</i>	<i>shi</i>	<i>tswanyi</i>	= <i>gbashitswanyi</i> (two hundred and forty thousand)
<i>gba</i>	<i>shi</i>	<i>twabà</i>	= <i>gbashitwabà</i> (two hundred and eighty thousand)
<i>gba</i>	<i>shi</i>	<i>tòtá</i>	= <i>gbashitòtá</i> (three hundred and twenty thousand)
<i>gba</i>	<i>shi</i>	<i>twani</i>	= <i>gbashitwani</i> (three hundred and sixty thousand)

When *kpákó* variant of *kpa* or *kpau* ‘two hundred’ becomes the root morpheme and prefix *egba* (*gba* or *gbau*) ‘two thousand’ affixed with the multipliers as suffixes, they will generate the following compound numbers:

7.

<i>gba</i>	<i>kpau</i>	<i>bà</i>	= <i>gbakpaubà</i> (eighty hundred thousand)
<i>gba</i>	<i>kpau</i>	<i>tá</i>	= <i>gbakpautá</i> (one million, two hundred thousand)
<i>gba</i>	<i>kpau</i>	<i>ni</i>	= <i>gbakpauni</i> (one million, six hundred thousand)
<i>gba</i>	<i>kpau</i>	<i>tsun</i>	= <i>gbakpautsun</i> (two million)
<i>gba</i>	<i>kpau</i>	<i>tswanyi</i>	= <i>gbakpautswanyi</i> (two million, four hundred thousand)
<i>gba</i>	<i>kpau</i>	<i>twabà</i>	= <i>gbakpautwabà</i> (two million, eight hundred thousand)
<i>gba</i>	<i>kpau</i>	<i>tòtá</i>	= <i>gbakpautòtá</i> (three million, two hundred thousand)
<i>gba</i>	<i>kpau</i>	<i>twani</i>	= <i>gbakpautwani</i> (three million, six hundred thousand)

Wubatacin (Infix)

This is the insertion of an infix into a word to change its meaning or value. This kind of affixation occurs in two occurrences in Nupe numeration. This is the insertion of an infix *tsu* into a numeric word to give its multiple. This can be explained as follows:

The number *kpautsun* ‘one thousand’ squares up when the derivational morpheme *tsu* is infixed into it as seen below:

8.

kpautsun ‘1,000’ - *kpau tsu tsun* = *kpautsutsun* ‘1 million’

This is expressed as *kpautsun* multiplied by *kpautsun* to give *kpautsutsun* ‘one million’; that is, one thousand multiplied by one thousand to give one million.

Also, the number *gbautsun* ‘ten thousand’ squares up when the derivational morpheme *tsu* is infixed into it as seen below:

9.

gbautsun ‘10,000’ - *gbau tsu tsun* = *gbautsutsun* ‘100 Million’

This is expressed as *gbautsun* multiplied by *gbautsun* to give *gbautsutsun* ‘one hundred million’; that is, ten thousand multiplied by ten thousand to give one hundred million.

Affixation in Nupe Ordinal Numerals

Affixation in Nupe ordinal numerals simply turns a cardinal number into an ordinal number with addition of the suffix /ci/ after a cardinal number. This is exemplified as follows:

Ordinal Numeral	Gloss
<i>nìní-ci</i>	1st
<i>gúbà-ci</i>	2nd
<i>gútá-ci</i>	3rd
<i>gúni-ci</i>	4th
<i>gútsun-ci</i>	5th
<i>gútswanyi-ci</i>	6th
<i>gútwabà-ci</i>	7th
<i>gútòtá-ci</i>	8th
<i>gútwani-ci</i>	9th
<i>gúwo-ci</i>	10 th
<i>gúwobègúbà-ci</i>	12 th
<i>gwegi-ci</i>	15 th
<i>eshi-ci</i>	20 th

<i>arata-ci</i>	50 th
<i>shitsun-ci</i>	100 th
<i>kpákó-ci</i>	200 th
<i>kpautá-ci</i>	600 th
<i>gba-ci</i>	2000 th
<i>gbautswanyi-ci</i>	12,000 th
<i>kpautsutsùn-ci</i>	1 millionth
<i>gbautsutsùn-ci</i>	100 millionth

Zubàgi (Adverbial Number)

These adverbs tell how often or how frequently an action takes place. They are formed when a cardinal number turns into an adverbial number with the affixing of the prefix /zu/ before it. It changes a noun or an adjective into an adverb. For instance, *gútá* is a cardinal number 'three', which is a noun or an adjective. When a prefix /zu/ is affixed to it, it changes to *zugútá*, meaning 'thrice' which is an adverb. See other examples below:

10.

- zu + níní = zuni/zuníní* (once)
- zu + gúbà = zugúbà* (twice)
- zu + gútá = zugútá* (thrice)
- zu + gúni = zugúni* (four times)
- zu + gútsun = zugútsun* (five times)
- zu + gútswanyi = zugútswanyi* (six times)
- zu + gútwabà = zugútwabà* (seven times)
- zu + gútótá = zugútótá* (eight times)
- zu + gútwani = zugútwani* (nine times)
- zu + gúwo = zugúwo* (ten times)
- zu + gúwo bè gúbà è = zugúwo bè gúbà è* (twelve times)
- zu + gwegi = zugwegi* (fifteen times)
- zu + eshi = zushi* (twenty times)
- zu + arata = zurata* (fifty times)
- zu + shitsun = zushitsun* (hundred times)
- zu + kpákó = zukpákó* (two hundred times)
- zu + kpautá = zukpautá* (six hundred times)
- zu + gba = zugba* (two thousand times)

Compounding

This is another morphological process observed in Nupe cardinal and ordinal numerals. Compounding is the morphological operation that generally puts together two free forms to derive a new word or numeral. The types of compounding observed in the cardinal numerals are the closed compound words, where two words are joined together without a space between them; and the open compounding, where words are joined together with a space between them. Nupe adverbial numerals (numerals that express countable number of times) are good examples.

The closed compound word or numeral involving putting two words together, without a space between them is exemplified in cardinal numerals in the data below:

11.

- i. *guwobènìní* 'eleven (11)'
10 plus 1
- ii. *guwobègubà* 'twelve (12)'
10 plus 2
- iii. *gwegidìngubà* 'thirteen (13)'
15 minus 2
- iv. *gwegidìnnìní* 'fourteen (14)'
15 minus 1

Closed compound numeral is also exemplified in ordinal numerals in the data below:

12.

- i. *guwobènìníci* 'eleventh (11th)'
10 plus 1
- ii. *guwobègubàci* 'twelfth (12th)'
10 plus 2
- iii. *gwegidìngubàci* 'thirteenth (13th)'
15 minus 2
- iv. *gwegidìnnìníci* 'fourteenth (14th)'
15 minus 1

On the other hand, the open compound word or numerals involve putting two words together, with a space between them, often expressing Nupe adverbial numerals (numerals that express countable number of times). These are exemplified as in the cardinal numeral in the data below:

13.

- i. *kpautsutsùn tú nìní* 'one million and one (1,000,001)'
one million and one
- ii. *kpautsutsùn tú gubà* 'one million and two (1,000,002)'
one million and two
- iii. *kpautsutsùn édá gubà* 'two million (2,000,000)'
one million times two
- iv. *kpautsutsùn édá guta* 'three million (3,000,000)'
one million times three

Open compound numeral is also exemplified in ordinal numerals in the data below:

14.

- i. *kpautsutsùn tú nìníci* 'one million and first (1,000,001st)'
one million and one
- ii. *kpautsutsùn tú gubàci* 'one million and second (1,000,002nd)'
one million and two
- iii. *kpautsutsùn édá gubàci* 'two millionth (2,000,000th)'
one million times two
- iv. *kpautsutsùn édá gutaci* 'three millionth (3,000,000th)'
one million times three

Compounding in these instances is achieved through a process of merging which means a process of packing of existing numerals together to indicate new quantity. This process plays a significant

role in the Nupe numeral derivation. It enables the counting to cover a large domain as well as counting of large quantities. The process is achieved through addition, subtraction and multiplication as well as the use of the morpheme *tu* ‘and’; where addition is indicated by the morpheme *bè* ‘plus’, subtraction is indicated by the morpheme *din* ‘minus’ and multiplication indicated by the morpheme *édá* ‘times’ respectively, as observed in the data (41 – 44).

Reduplication

Reduplication is another morphological process observed in Nupe cardinal and ordinal numerals. Reduplication is a word formation process where a word or part of a word is repeated to create a new word with a new meaning. This can be full reduplication (repeating the entire word) or partial reduplication (repeating only a part of the word). Nupe cardinal and ordinal numerals exhibit partial reduplication where only part of the base is repeated. Partial reduplication in Nupe cardinal numeral is exemplified in the data below:

15.

- i. *kpautsùn* ‘1,000’ - *kpautsu-tsùn* = *kpautsutsùn* ‘1million’
- ii. *gbautsùn* ‘10,000’ - *gbautsu-tsùn* = *gbautsutsùn* ‘100 million’

Partial reduplication is also exemplified as in Nupe ordinal numerals in the data below:

16.

- i. *kpautsùnci* ‘1,000th’ - *kpautsu-tsùnci* = *kpautsutsùnci* ‘1 millionth’
- ii. *gbautsùnci* ‘10,000’ - *gbautsu-tsùnci* = *gbautsutsùnci* ‘100 millionth’

Agglutination

This is a fundamental morphological feature of the Nupe language and numerical system. It is a morphological condition where words are formed by stringing together morphemes (word parts), each of which corresponds to a single syntactic feature. Morphemes in Nupe numeral combined in the sentential derivation are separable and they retain their distinct meanings as seen in the data below:

17.

- i. *kpautsutsùn tú nìní* ‘one million and one (1,000,001)’
one million and one
- ii. *kpautsutsùn tú gubà* ‘one million and two (1,000,002)’
one million and two

Discussion of Findings

This study explored the morphological derivation and features of the Nupe numerals. It established that Nupe numerical system utilizes affixation, compounding and reduplication strategies in deriving the numerals. The study also found that agglutination is also a fundamental morphological feature of the Nupe numerals. This further confirms that the language is an agglutinative language since morphemes combined in the sentential derivation are separable and they retain their distinct meanings. The study enriches the knowledge of the linguistic features of Nupe numerals and informs learners, linguists and researchers on the morphological processes involved in deriving them. The findings of this study contribute to the existing knowledge of Nupe language and linguistics. This work is of great importance to both the native speakers and the second language learners as it will aid them in improving their counting skills. It will immensely aid the native and

the second language users in improving their language usage or skills. The findings also serve as a contribution to the existing body of knowledge and few researches conducted in the area of Nupe numerical system.

Conclusion and Recommendations

It is indeed a truism that every language has its peculiar counting or numeral system, and so does the Nupe language. In the Nupe numerical system, numbers are derived through the use of additive, subtractive and multiplication strategies manifested through morphological processes such as affixation, compounding and reduplication. Therefore, the study concludes that adequate knowledge and understanding of these morphological processes is imperative for competence and proficiency in the Nupe counting system. However, it is clear that the traditional Nupe numeral system is under threat of extinction, due to lack of documentation coupled with the domineering influence of the English language over the Nigerian languages. Hence, linguistic documentations such as this are imperative to analyze and document the traditional patterns before they succumb to the global pressure to westernization.

Based on the conclusion, the study recommends further research on Nupe numeral system and language considering its level of description, while efforts should be made to record and document traditional counting systems of minority languages. Also, Nupe speakers should try to preserve the language by speaking it and encouraging their children to acquire Nupe language as their first language, as this cannot stop them from becoming fluent in English. Likewise, cultural renaissance among native speakers on the importance of indigenous languages coupled with the encouragement and training of teachers in indigenous languages at all levels of education from the primary, secondary to tertiary institutions should be encouraged.

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